

Chapter (15) + Chapter (16)

Query Processing and Query Optimization

I. Consider the following relations:

student (ID, name, tot_cred)

takes (ID, course_id, semester, year, grade)

course (course_id, title, dept_name, credits)

Consider the following relational algebra expression:

$$Q = \pi_{name, title} (\sigma_{dept_name='CS' \wedge credits=3} (student \bowtie takes \bowtie course))$$

The following statistics are available: the student contains 10,000 tuples and 1,000 blocks, takes contains 30,000 tuples and 3,000 blocks, and the course contains 2,000 tuples and 500 blocks. In addition,

- $V(dept_name, course) = 50$
- $V(ID, student) = 10,000$
- $V(ID, takes) = 10,000$
- $V(course_id, takes) = 2,000$
- $V(course_id, course) = 2,000$

Assume that 20% of the courses have credits = 3. The block transfer time is ($t_T = 0.1$) ms and the disk seek time is ($t_S = 4$) ms. Assume that the number of disk seeks for a sequential scan is equal to the number of relations scanned, and that the estimated number of blocks is equal to the number of block transfers.

- Transform the given relational algebra expression into an optimized relational algebra expression.
- Draw the query trees for the original and optimized query plans.
- Estimate the number of tuples in each intermediate result for both query plans.
- Estimate the number of blocks in each intermediate result.
- Calculate the total cost of both query plans.
- Determine which query plan is more efficient and calculate how many milliseconds are saved by using the optimized query plan.